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Simple and Multiple Linear Regression

CHAPTER OBJECTIVES
■ Learn and appreciate why regression analysis is so central to virtually all statistical models and how many models are simply extensions of the fundamental regression model. ■ Understand the nature of least-squares regression, what it accomplishes, and even more importantly, what it cannot do. ■ How to estimate a regression equation when you have one response variable and one or more predictors. ■ Decipher and distinguish between R^2 and adjusted R^2, and appreciate the benefits of the latter over the former. ■ Understand how multiple regression differs from simple linear regression, and the dangers of not fully appreciating this distinction. ■ How to run and interpret simple linear regression and multiple regression analysis in Python. ■ Learn model-building strategies such as forward and stepwise regression, and critically evaluate each as a way of performing regression. ■ Overview of mediation analysis in statistics, and why conclusions drawn from such analyses are often incorrect and overstep what the data actually can say.

Many statistical techniques used by researchers, in one way or another, even if somewhat indirectly, boil down to an enhanced version of **regression analysis**. Just as techniques such as *t*-tests for means could be conceptualized as special cases of ANOVA, the analysis of variance itself can be considered a special case of the wider regression model framework. Historically, regression analysis arose in the late 1800s, while ANOVA was innovated around the 1920s. And actually, the historical seeds of both techniques were laid much earlier at the turn of the nineteenth century (i.e. early 1800s) by Gauss and Legendre (Stigler, 1986). Though on an applied level they arose for different purposes, on a technical level, as we will see, they are remarkably similar. If you are a newcomer to statistics and research, you may at first glance be surprised to learn that there are literally **books and volumes dedicated solely to regression analysis**. One reason for this is because many, or perhaps even most of the topics and

issues that arise in regression are also prevalent in other statistical models as well. And if you are very familiar with these topics in regression, you will not see them as entirely “new” when encountering other statistical models, even the so-called relatively “advanced” ones. Sometimes researchers, using a “modern” statistical method (e.g. in machine learning, for instance), discuss issues with the method that they mistakenly believe are universal only to that method, without realizing that the issue with the “advanced” statistical method is really a page out of a first-year introductory regression book! If one is unaware of the **foundations** on which what they are doing are based, they are more likely to believe in the “hype” of new statistical methods, without realizing that much of the new methods may consist of ingredients simply “glued together” from other methods. Now, that is not to say new methods are not rightfully “new.” They are, and innovators of newer methods are remarkable scholars. It is only to say that the concepts inherent in new methods usually have roots in prior methods. A failure to understand or appreciate this leads one to believe that new concepts (as opposed to the specific technique) are being “invented” with new methods. While this is sometimes true, it is usually not the case.

The dependent variable in regression is often called a “**response**” variable, in the sense that it is “responding” to the value entered by the predictor variable. The **predictor** variable in regression is the corresponding independent variable in ANOVA. It really does not matter which terms you use, but “independent variable” is more times associated with **manipulation**, such as one might have in an experimental context, whereas “predictor variable” is a bit more general in scope. The idea of manipulating an independent variable comes from the mathematical literature, where the function statement $f(x)$ implies that as you purposely change the value for x , the value of the function will be affected (i.e. it will also usually change, unless it is, for instance, an identity function). Hence, when we change the values of the independent variable, the “response” (which in this case are the function values, or “ y ”) will also change. Even in ANOVA settings researchers may speak of predictor variables even when experimental control and variable manipulation is at its highest. **Terminology in statistics or mathematics is not standard, and often you have to read the situation and decipher what is being communicated.** This sometimes makes learning from a new textbook, for instance, mostly an exercise in understanding what notation is being adopted! As Feynman once remarked, “We could, of course, use any notation we want; do not laugh at notations; invent them, they are powerful. In fact, mathematics is, to a large extent, invention of better notations.” Feynman meant this more in the spirit that similar concepts can be found and communicated in different branches of mathematics (i.e. mathematics is filled with equivalences across domains), but the point is that to understand a definition or new term in a given field, you have to ask yourself what the concept actually “means” and whether it has been represented by something else in the past or in an allied field.

When more than a single predictor is entered into the regression simultaneously, but still with only a single response variable, the model is that of **multiple linear regression**. Be sure to note we still have only a single response variable in this model even if we have more than a single predictor. If, on the other hand, the model has several response variables and several predictor variables, the model is that of **multivariate multiple linear regression**. Multivariate statistics is the topic of later chapters in the book. These models require some justification for entering several response

variables into the same model. As we will discuss when we survey the multivariate analysis of variance, or “MANOVA” for short, the set of response variables entered into the model should make up what is known as a **variate** in statistics, which is, for now, a meaningful, or at least substantively “plausible,” **linear combination** of variables. As mentioned, we delay our discussion of multivariate models until later in the book.

7.1 Why Use Regression?

As always, before using a statistical technique, one must first be aware of why one would even want to use it. Often students and researchers use a statistical method, thinking it accomplishes what they have in mind for it, yet not realizing the limitations of what the statistical procedure can actually provide. This is especially true in the case of regression. If the statistical method is used without understanding what actually goes on “behind the scenes,” then when interpreting the output of the method a researcher can quite easily make philosophically weak or untrue conclusions of what the actual knowledge is that is provided by the procedure. “Behind the scenes” is what the procedure is actually doing! The **bare bones mathematics and philosophy**. The rest are just words and potential misunderstandings of what a researcher would “like” the method to be doing. These are two different things, and the distinction between them is very important to understand. The bridge between what a statistical method actually tells you vs. what you would **like** it to tell you on a scientific basis is not a philosophically simple leap.

Let us feature an example to demonstrate what we mean. Suppose a researcher specializes in the study of alcoholism and is interested in knowing whether drinking behavior is a predictor of suicide. That is, the researcher would like to know whether, in general, the more one drinks, the higher the risk of suicide. Hence, the researcher frames the question this way:

Is there evidence that alcoholism predicts suicide?

The key word in this is, of course, “predicts.” What does the word mean? Now, at first glance, you might think to yourself, “That’s a silly question. Everyone knows what predicts means.” While that may be true in an ordinary and everyday language sense of how the word is used, in statistics and science, ordinary “English” will not suffice. **In statistics, the word “predict” only means as much as how it is defined mathematically.** Otherwise, it has no meaning. Hence, if the researcher runs the regression by having alcoholism as the predictor variable and suicidality as the response, and then draws a conclusion of “alcoholism predicts suicide,” the statement is hollow for that researcher unless there is an understanding of how the statistical procedure that had been run operationally defines “predict.” Otherwise, the researcher is simply running a regression analysis without even understanding what the procedure even does beyond associating words to it such as “predict!” What is more, if the audience is not aware of what “predict” means on a deeper level than some arbitrary semantic meaning, very little in the way of true communication is happening. “Alcoholism predicts suicide” and nobody in the room knows what it means! Insanity! The software does not know either, as it is just doing its job at computing. To understand the words you use in a statistical context, you need to have at least some appreciation for what the statistical methodology is doing “behind the scenes,” otherwise the communication is

hollow. You may still get the publication, but you will not appreciate what you yourself are even publishing. This is why understanding statistical methods for scientists is extremely important. As discussed earlier in the book, this is one danger of associating statistics and research solely with software analyses. You have to understand what the software is doing, otherwise, run the regression as you may, you will be at a loss to understand its output at any level other than associating imprecise **verbal labels** to what you are reading. The good news is that for topics like regression, and many other statistical techniques, understanding what the procedure does and how it works does not often require advanced training. What it does require, however, is careful study of the method's principles along with a discussion of what can vs. what cannot be concluded from the method. As highlighted throughout the book, far too many statistical methods are used to draw research conclusions that are simply not warranted. Understanding the statistical method allows one to better appreciate the limitations of what one has computed, and be in a better position for interpreting the research result accurately.

The following are a few very powerful and recent samples and examples of why understanding the statistical tool is so important and vital to good interpretation of the ensuing scientific claim:

- In the COVID-19 pandemic, it was often said by scholars in the area as well as media types that **models are only as good as the assumptions that go into them**. That is, one can make predictions of COVID-19 deaths as much as one would like, but unless the assumptions and other a priori “inputs” to the model are reasonable, then the predictions will necessarily be faulty. We heard this repeatedly on television. If someone uses regression, or any other statistical model, yet does not understand how influential assumptions can be to the model, then they will necessarily misinterpret the findings of the model and mislead those they publicize the model to.
- In the death of **George Floyd** in 2020, the question was raised as to whether being black, in general, is predictive of mistreatment by police, now and historically. Papers examining and seeking to demonstrate a predictive relationship will likely be using (and have used in the past) a statistical model (often a version of regression) to do it. Hence, when a conclusion is made, “Black citizens are more likely to be victim of police brutality than are white citizens,” it is imperative that the consumer of such research be able to disentangle what is meant by “predictive” in this case. What does it mean, precisely, in the sense it was used in the regression? If the researcher or consumer of such research is unaware of the underlying principles that govern the statistical model that presumably led to the conclusion (and how those elements could be adjusted on a technical level), then that researcher will be unable to critically evaluate whether the conclusion is correct or incorrect. That is, you will be unable to evaluate the claim that is being made. **If you want to be able to evaluate the claim, you need to understand what the statistical method is doing.** There is no way around this work. What does an educated and informed conclusion in this regard sound like? Something like this: “There is evidence that color predicts police mistreatment ... in the least-squares sense where squared errors in prediction are minimized.” This is an example of a conclusion interpreted by someone who understands how a statistical model can impact the conclusion. As a scientist, you should be aiming for making, understanding, and paraphrasing such conclusions if you are to be an intelligent consumer and producer of research.

- Is a poll conducted three months prior to a presidential election predictive of the final election results? As we saw in 2016, most polls were not predictive (well, they were predictive, only of a different result). Understanding polling and the statistics on which the poll is constructed is much more complicated than the succinct conclusions drawn from television personalities. You have to understand what “predictive” means in this sense, and that knowledge comes from knowing how the statistical model works and what conclusions can be drawn from it. **If you want to know what the microscope is detecting, you need to understand a bit about the microscope, not only that which you are seeking to observe.**

Hence, it may sound like an obvious statement, but it is imperative to understand how regression predicts responses if one is to use the word “predict” after performing a regression on some data! Otherwise, it is all “shirt and tie, and no substance.” Again, the good news is that in depth mathematics are not absolutely required to understand the most important features of what makes regression “tick” and all the assumptions that go along with it. Now, that does not mean that more mathematics or philosophical justification is not always a good idea. For instance, knowing more of the “underground” of **matrix algebra** will only lead to more enlightenment down the road, just as understanding the logical precursors that must be in place before prediction is entertained for a given research problem will foster a richer understanding. Mathematically, knowing how to pump out derivatives and integrals, for instance, is not nearly as important as understanding how regression works on an intuitive level, while also not ignoring the technicalities.

7.2 The Least-Squares Principle

Though regression analysis may use one of several methods of estimating parameters, by far the one that dominates is **ordinary least-squares**. In fact, much of the history of regression analysis dates back to the origins of the least-squares principle in one form or another. Many early astronomers, for instance, worked on ways of minimizing error in predicting astronomic phenomena (Stigler, 1986). Though the mathematics behind least-squares can get rather complex, the principle is very easy to understand. When we say “least-squares,” what we mean is minimizing (i.e. “least”) a quantity. The quantity we are wanting to minimize is a squared quantity (i.e. the “squares” part of the principle). But, what quantity do we want to minimize when making predictions? Remarkably, it is analogous to making predictions without least-squares, and we have already seen it before. If you follow our development here, you will have a solid and intuitive understanding of least-squares.

Recall the sample variance discussed earlier in the book:

$$s^2 = \frac{\sum_{i=1}^n (y_i - \bar{y})^2}{n-1}$$

In computing the variance, recall that, in the numerator, we are summing squared deviations from the mean, but it is entirely reasonable to ask **why we are summing deviations from the mean, and not any other number?** Why we are squaring deviations is well understood, so that the sum does not always equal 0 (i.e. if we did not

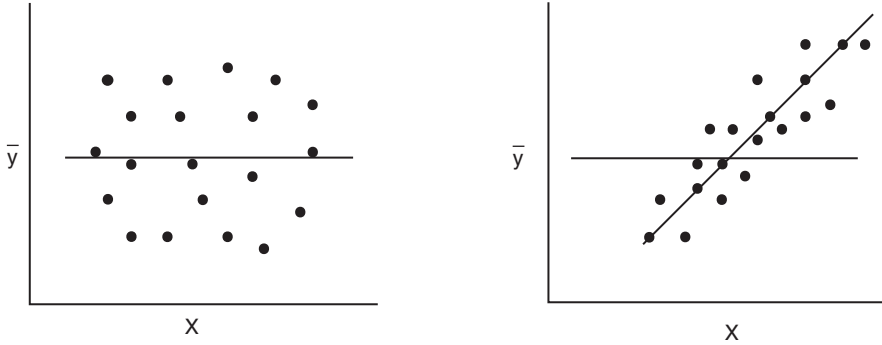
square deviations, then $\sum_{i=1}^n (y_i - \bar{y})$ would always equal 0 regardless of how much variability we have in our data; try it yourself with some sample data), but why deviations from the **mean** in particular? Believe it or not, the answer to this question helps form the basis of least-squares regression, and the answer is this: because **taking squared deviations from the mean ensures for us a smaller average squared deviation than if we took deviations from any other value, such as the median or the mode** (Hays, 1994). That is, taking a sum of squared deviations minimizes the squared error. The theory behind this result is beyond the scope of this book (see Hays, 1994, for details), but suffice it to say it can be quite easily justified and is on a solid footing. The important point is this, that the variance generates a quantity that minimizes the squared error in prediction. Hence, if you guess the mean from a set of data, you are guaranteed to experience the lowest average squared error possible when you make predictions using this principle. The variance, in effect, provides a least-squares solution, but under the condition of either no predictor, or, if there is a predictor in the model, that the correlation between the response and the predictor is equal to 0. This important conceptualization of the variance paves the way to understanding least-squares as it is applied to ordinary regression analysis, which we now discuss and develop.

7.3 Regression as a “New” Least-Squares Line

To understand regression, it is first helpful to start with the case where there is zero linear correlation between the predictor and the response. If such were the case, then in order to minimize error in prediction overall, the best value to predict would be the **arithmetic mean**. Hence, if the correlation is equal to 0, or there is no predictor at all, the best prediction for the response in the sense of keeping errors to a minimum would be the mean of the response variable. Even under this circumstance, we would want to know how much error we could expect if we made predictions repeatedly using the average. In other words, we need a measure by which to index the degree of error we would, on average, expect to experience if we were to use the mean as a predictor. The answer to this question is the **standard deviation**, which recall is the square root of the variance. That is, under the case of zero correlation, our best prediction would be the mean, and the degree to which we could expect to be “off” in our prediction can be indexed by the standard deviation of the response variable. However, if we do have a correlation between the predictor and the response, then it follows that if we were to use the standard deviation as our error in prediction estimate, it would not adequately account for the average error in prediction. We could still use it as our estimate, but it would not be nearly as useful as if we measured error around a “new” regression line, which is the least-squares line we will fit to our data.

Consider the following two plots. The first plot (left) is one where there is zero correlation between the predictor and the response. In such a case, the arithmetic mean is our best predicted value. In the second plot (right) is where there is a correlation. Consequently, the mean is no longer our best predicted value and instead of using a horizontal line in our prediction, we will “tilt” the line upward. This new line is our regression line, and the degree of error around the line will be indexed by the standard deviation of observations around the regression line. This standard deviation of scores

around the regression line has a special name, which is called the **standard error of the estimate**, and is an estimate of the population standard deviation of scores around the population regression line. Hence, we can see that the least-squares regression line constitutes a “tilting” of the original line where the arithmetic mean was our best prediction under the condition of zero correlation.



7.4 The Population Least-Squares Regression Line

To obtain the tilted line, we need to know first and foremost how to compute a line. As you may recall from high school mathematics, a line is made up of an **intercept** and a **slope**:

$$y_i = \alpha + \beta x_i$$

where y_i is the observed value along the line, α the intercept of the line, β the slope, and x_i is the value for the predictor variable that we are inputting into the equation to compute the function and obtain y_i . Now, if we were doing mathematics, this would be the end of the story. That is, if we were not fitting the line to a messy swarm of data, the above equation is all we would need. **This is where statistics differs from mathematics.** In statistics, we are fitting the line to a messy swarm of data, and hence, in addition to the linear function above, we need to append a quantity that indicates the reality that in most of our predictions using the line, we will be “off” by a certain degree. This chance of “error” is designated as ε_i and represents the difference between the observed value in our data and the predicted value using the linear function. When we append the error term to our equation, we have

$$y_i = \alpha + \beta x_i + \varepsilon_i$$

This is the **population regression line**. It assumes we have the actual population data, which we typically will never have, and hence we will have to estimate the parameters α and β using sample data. The parameter ε_i , as mentioned, is the difference between observed and predicted values, equal to $y_i - \hat{y}_i$, where $\hat{y}_i$ is the predicted value for a given input of x_i into the equation. It is estimated as well, though we are much more interested in estimating α and β . Note that α and β are not indexed by i . This is because, for any given population, they are considered **constant**, in that they

are the only intercept and slope we are fitting for the given population model. The response variable, y_i , along with the error term, ε_i , are indexed by i to indicate that there will be a different value for each value of x_i entered into the equation. Hence, we can say that the first part of the equation, $\alpha + \beta x_i$, is the **deterministic** component of the model, while the second part, ε_i , is the more **random element** of the model. This random portion of the model is sometimes referred to as the **stochastic** portion of the model. Hence, we see that the regression line, at least in structure, is no different than a regular line from elementary mathematics, only that it now includes an error term to account for being fit to messy data. That is, we can make predictions for y_i , but we can expect our predictions to be off to some degree, which is indexed by ε_i . The next question is how to compute sample estimates for α and β so we can use the equation for prediction. We consider that problem next.

7.5 How to Estimate Parameters in Regression

What we have defined thus far is the population regression line, but we have not discussed how to estimate parameters that make up the equation. Why estimate parameters? Well, if we actually had the true population regression line, we would have no need to estimate anything. For example, if I deemed my population to be equal to one hundred data points, computed the regression line on such data, and were not interested in making inferences toward a larger set of observations (obviously, since I have already deemed my regression equation to be that of the population), then we would have no need to estimate parameters. However, as usual in science and statistics, we are rarely if ever working with population data. Instead, we are working with sample data. Hence, values of α and β will be considered as **unknowns** and will have to be estimated from sample data. What we need now are equations that will tell us exactly how to compute the **sample estimators**.

The estimators for α and β will be denoted by $a_{y \cdot x}$ and $b_{y \cdot x}$, respectively, where the subscript notation $y \cdot x$ indicates that we are predicting y based on x . So, how do we compute these sample quantities? Though the derivation on why they are computed the way they are is well beyond the scope of this book, the most important thing to recognize for our purposes is that these quantities must be computed in a very particular and exact way. Hence, when we show you the formulas for computing them, from an applied perspective, it is enough to realize that these formulas were developed via **differential calculus** so that they ensured a particular condition was satisfied. What condition do they guarantee? They guarantee that if we compute them in this way, we can be assured that the **sum of squared errors around the regression line will be smaller than anywhere else we may have chosen to fit the line**. This is the least-squares principle we discussed earlier. These are called the **least-squares solutions** for this reason. If you compute them in this way, you get the **least-squares regression line**. If you compute them in any other way, you do not. Since we wish to minimize the sum of squared errors around the line, it is essential then that we use these particular formulas. In other words, this is why these formulas “are the way they are,” because they ensure we are minimizing a function, in this case, the least-squares function. In general, many formulas are the way they are in statistics because they **minimize** or **maximize** a function. They derive from topics in a field of mathematics called **optimization** where functions are maximized or minimized, given that particular

and well-defined constraints are in place during the process. Keep that in mind when you see formulas and ask yourself why the given formula is the way it is. It is usually the way it is for a very good reason, which is often one of optimization criteria.

The **intercept estimator** $a_{y.x}$ is computed by

$$a_{y.x} = \bar{y} - b_{y.x}\bar{x}$$

while the **slope estimator** is computed by

$$b_{y.x} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Again, these are referred to as the least-squares solutions for the regression line. If you compute these equations on sample data, you have obtained intercept and slope values that will ensure that when you fit them to data, you have minimized the sum of squared errors around the regression line. That is, you will be guaranteed to have minimized

$$\sum_{i=1}^n e_i^2 = \sum_{i=1}^n [y_i - (a + bx_i)]^2$$

which is the unpacked version of $\sum_{i=1}^n e_i^2 = \sum_{i=1}^n [y_i - y_i']^2$ since $y_i' = a + bx_i$, and the **residual error** is equal to “observed minus predicted,” or $y_i - y_i'$. Without delving too much into the details of each equation, we can nonetheless get a good feel for what each term “does” in terms of its computation. For instance, look at the equation for the intercept: $a_{y.x} = \bar{y} - b_{y.x}\bar{x}$. Notice we are starting out with the sample mean, $\bar{y}$, and subtracting out $b_{y.x}\bar{x}$. Why start out with the sample mean in the computation? We start out with $\bar{y}$ because under the condition of zero correlation between x_i and y_i , it follows that $b_{y.x}$ will also equal zero. This makes sense, since if you picture a flat slope, this also indicates, as already discussed, that the linear correlation between the two variables is also zero. Therefore, when $b_{y.x} = 0$, then it follows that

$$\begin{aligned} a_{y.x} &= \bar{y} - b_{y.x}\bar{x} \\ &= \bar{y} - (0)\bar{x} \\ &= \bar{y} \end{aligned}$$

On the other hand, when $b_{y.x} \neq 0$, then $a_{y.x} \neq \bar{y}$, which reflects the fact that the line has been “titled” to better fit the observed data. Since $b_{y.x}$ may be greater or less than 0, this will also be reflected in the computation for $a_{y.x}$. We can appreciate then why $a_{y.x}$ is the way it is, even if we have not proved it mathematically. Only a mathematical proof provides ultimate justification. What we have provided is but a breakdown of why the formula might make sense to us on an intuitive level.

How about $b_{y.x}$? If you look at this formula a bit more closely, how it is computed will also make good sense. Again, we are not justifying or proving the formula. Instead, we are looking to see what the formula is “communicating” to us. What is it trying to tell us? We will now look at it in more detail:

$$b_{y.x} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

We can see that in the numerator is a sum of product deviations, $\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$, while in the denominator is a sum of squares for the predictor. The numerator incorporates the response variable, while the denominator does not. The extent to which the numerator is larger than the denominator is the extent to which $b_{y,x}$ will be different from 1.0. Notice that $b_{y,x}$ is actually communicating a ratio of **cross-products**, since we can unpack $\sum_{i=1}^n (x_i - \bar{x})^2$ into $\sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})$, and hence we rewrite $b_{y,x}$ as

$$b_{y,x} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})}$$

If both sums are the same, then the slope is 1.0. If the amount of cross-product sum in the numerator is equal to 0, then this implies

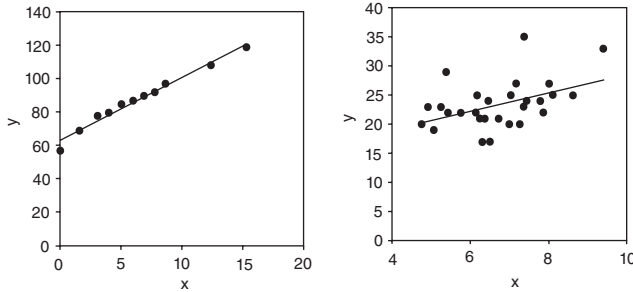
$$b_{y,x} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{0}{\sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})} = 0$$

regardless of how much variation is in the denominator. Of course, if there is no variation in the predictor variable, then $\sum_{i=1}^n (x_i - \bar{x})^2 = 0$ and since division by 0 is **undefined** (not “zero,” but rather undefined), the slope in this case would also be undefined. The formula for $b_{y,x}$ is literally answering the question of how big one cross-product is in the numerator relative to another cross-product in the denominator. The cross-product in the numerator incorporates both y_i and x_i ; the cross-product in the denominator only features x_i .

In summary, then, least-squares regression theory ensures that if we compute the intercept and slope by the above equations, we will, on average, minimize errors in prediction when using x_i to predict y_i . If you understand that, you now have a pretty good grasp of what “prediction” means, in the least-squares “sense” of the word. Estimation of parameters can also be performed using a technique called **gradient descent**, which can be especially useful for large-scale “big data” problems. For most common data analyses, least-squares, or a derivative of it, will suffice. For details on gradient descent, consult Bishop (2006).

7.6 How to Assess Goodness of Fit?

Once we have fit our regression line using the least-squares principle, the question that needs to be answered next is how well does it fit? While a least-squares line is guaranteed to minimize the sum of squared errors, simply minimizing a function does not, by itself, guarantee a good fit to the data. By analogy, a football team might work hard to minimize making errors on the field, but this does not in any way ensure the football team will be a good team. How “good” the fit of the regression line is, is unrelated to the minimization criteria imposed on it to compute the least-squares estimates. As an example, consider the following two plots of bivariate points:



In both cases, the least-squares line is accomplishing the same thing, that of minimizing error. However, the fit of the regression lines to the data in each case could not be more different! Notice in the first plot (left), the fit is quite strong around the regression line. In the second plot (right), the fit is quite poor. What, then, will determine how well a least-squares line fits the data? This is not a statistical issue, it is a scientific one. **The quality of data and the strength of the relationship is what will determine the fit, nothing else.** In both plots, the regression lines are accomplishing the same task, that is, they are both **minimizing the sum of squared errors**, and are each doing it equally well! What differentiates the plots is the degree to which the data conform to the regression line. Hence, what we need is a measure of **model fit** or equivalently, a measure of the **average error** around the regression line. There are many measures we can use to assess model fit. We will survey a couple of them now.

7.7 R^2 – Coefficient of Determination

Undoubtedly, the most popular measure of model fit in a regression is that of R^2 , often called the **coefficient of determination**, for the reason that it measures the variance in the response variable accounted for or “explained” by knowledge of the predictor variable, or, in the case of multiple linear regression, predictor variables. R^2 is defined as

$$R^2 = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

A look at R^2 helps us appreciate what it is accomplishing. In the numerator, notice we are taking squared deviations of predicted values from the mean value of the response, $\sum_{i=1}^n (\hat{y}_i - \bar{y})^2$. This is contrasted in the denominator to simply a sum of squared deviations of observed values from the mean of the response. Hence, what differentiates the numerator from the denominator is whether we use $\hat{y}_i$ or y_i when subtracting the mean, $\bar{y}$. The extent to which the two values agree will be the extent to which R^2 is equal to 1.0. On the other hand, the extent to which the numerator accounts for lesser variation is the extent to which R^2 (in the limit) will approach 0. Notice that, typically, R^2 cannot be negative in value, since both the numerator and denominator must be positive quantities.

Having defined R^2 , what does it tell us on a scientific level? That is, we know statistically how it is defined, but how does that translate into drawing scientific conclusions?

The truth is that when we say R^2 accounts for “variance explained,” this is all it accomplishes. It does not claim to do anything more than that. Like the correlation coefficient, it cannot pretend to know the nature of your variables or whether there were any experimental controls implemented in your research. R^2 , like any other statistic, is an **abstract measure** applied to data and hence the fact remains that if you want to draw strong conclusions from your research, the design of your research is still paramount. **Statistics do not save you from poor design!** And so, even if we obtained an R^2 near 1.0, any kind of directional force or “impact” of one variable onto another cannot be inferred by this figure alone. One needs a good research design in order to draw such claims (and even then, it is usually a methodological challenge).

Conveniently, R^2 can also be defined as the **squared correlation between observed and predicted values from the model**. If you think about what this means for a moment, it should make good sense. Recall that the purpose of a statistical model is to explain or account for variability. If the model does this well, then it should be “reproducing” the data, or, at a minimum, **regenerating a linear transformation of the data**. If predicted values align perfectly with observed values, the model, from a statistical point of view at least, is a definite “winner.” Hence, the extent to which observed and predicted values line up is the extent to which the model fits the data. And so, this interpretation of R^2 is quite convenient and useful as well. The goal of precise mathematical modeling is generally to see how well a function rule can be obtained that, in a strong sense, duplicates the observed data. The function rule can be thought of as a “narrative” for “explaining” the observed data. Function rules that do this well fit the data very well. The coefficient of determination is a good way to first assess how well that narrative fits. It does not imply it is the “only” narrative that could have explained the data. It only means that the current narrative does a nice job. In theory, one can have an infinite number of narratives explain the same data, something sometimes researchers fail to acknowledge in reporting their model as if it is “the” only model that could explain the observed data. Be on the lookout for researchers who buy into their theory a bit too much. Many other models or narratives may fit the data just as well. Especially with social research, **one narrative that fits is likely one of many that could fit**.

7.8 Adjusted R^2

Though R^2 is a useful descriptive statistic of how much variance is being accounted for in the sample, it has a pitfall in that it is **biased upward** as an estimator of the population effect size. That is, R^2 will more times than not lead to an overly “optimistic” estimate of the true population effect size. This will be true especially for small samples. For larger samples, the bias is much less, and eventually disappears as sample size grows without bound. What is more, R^2 **can typically only increase as one adds more predictors to a model, even if the contribution of those newly added predictors is minimal**. This is something extremely important to understand. Hence, if I have a regression model with three predictors with a given explained variance and then add a fourth predictor, R^2 will either stay the same or increase. It cannot decrease.

So why is this a problem? The problem is not so much statistical as it is, once again, scientific. In short, it means that **one can continually add predictive power to one's model by simply including more predictors**. But surely a model with 1,000 predictors, as an extreme example, is not very useful on a scientific level. If we can explain nearly as much of the variance with, say, 10 predictors, then the more **parsimonious** model must be preferred. Adjusted R^2 tackles this issue of parsimony head-on, rewarding models that are more parsimonious than not. Consider how it does this through a look at its formulation:

$$R_{\text{Adj}}^2 = 1 - (1 - R^2) \left(\frac{n-1}{n-p} \right)$$

where n is the number of data points, while p is the number of parameters fit by the regression. This includes the intercept term. Hence, in a model with one predictor, p will equal 2. The “active” or “working” component in R_{Adj}^2 is thus the ratio $(n-1)/(n-p)$. Notice that for constant n , as p increases, the denominator $n-p$ gets smaller, and hence the ratio $(n-1)/(n-p)$ gets larger. How does this affect the size of R^2 ? In the formula, we will have to take the product $(1 - R^2)((n-1)/(n-p))$ first before subtracting, due to the order of operations (i.e. multiplication comes before subtraction). The value of $(1 - R^2)$ is the amount of variance **unexplained**, since it is the balance of R^2 . Therefore, the larger $(1 - R^2)((n-1)/(n-p))$ is, the more we are subtracting from the “best fit” value for R^2 that is represented by the starting point of the formula, 1.0. Hence, what will keep R_{Adj}^2 larger than not is if we are subtracting a smaller value than not for $(1 - R^2)((n-1)/(n-p))$. But, as p gets larger and larger, $(n-1)/(n-p)$ also gets larger, which means that $(1 - R^2)((n-1)/(n-p))$ gets larger, and hence, all else equal, larger p given a more or less constant original R^2 gives us a smaller R_{Adj}^2 when we subtract from 1.

Now, at first glance, based on our argument above, you may come to the tentative conclusion that increasing the number of parameters is a “bad thing” overall, since it seems to have a negative impact on the size of R_{Adj}^2 . However, this is not entirely true, and therein lies the genius of R_{Adj}^2 . If the new parameters increase R^2 by a relatively substantial amount, the increase in explained variance will be “worth” adding the new parameters. This is how R_{Adj}^2 can be considered a **penalized** version of R^2 , because if you add new predictors that are not worth it, **you get punished**. Hence, models are rewarded that are more **parsimonious** than not, which accords well with how science should work. This is how R_{Adj}^2 is of paramount importance from a **scientific** perspective, as well as statistical. **Parsimony is what science is all about**. Simple explanations are always considered better than more complex ones if they provide equivalent explanatory power. If you start adding every predictor under the sun with the hopes of simply increasing variance explained, R_{Adj}^2 says, “Nope, not on my watch,” and will have the effect of decreasing by an increasing amount the original R^2 . R_{Adj}^2 is not the only penalized statistic in regression. **Akaike's information criterion** (AIC) is another such measure that attempts to reward more parsimonious models over less parsimonious ones. We do not discuss AIC here. Interested readers are encouraged to consult James et al. (2013) for a good introduction.

7.9 Regression in Python

As an example of regression in Python, and at the same time to demonstrate a data management task, we will import a data file originally saved in SPSS. The original data file is named `iq_data.sav`. We can use `pyreadstat` to read it into Python. We will name the new data file by the name of `df`. We print out only a few cases of the file:

```

pip install pyreadstat
import pyreadstat
df, meta = pyreadstat.read_sav("iq_data.sav")
df
Out[321]:

```

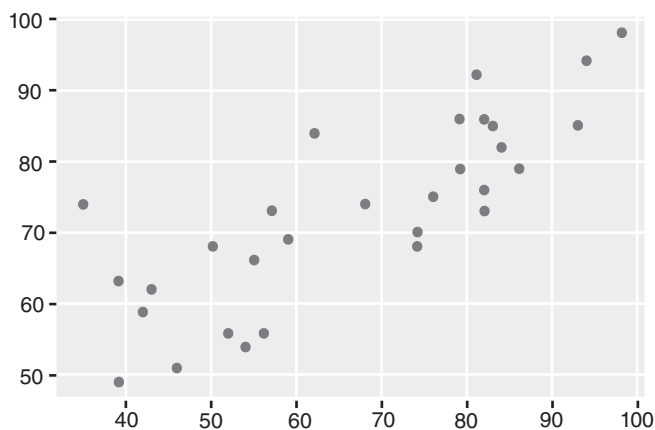
	verbal	quant	analytic	group
0	56.0	56.0	59.0	0.0
1	59.0	42.0	54.0	0.0
2	62.0	43.0	52.0	0.0
3	74.0	35.0	46.0	0.0
4	63.0	39.0	49.0	0.0
5	68.0	50.0	36.0	0.0
6	54.0	54.0	29.0	0.0
7	56.0	52.0	57.0	0.0
8	51.0	46.0	65.0	0.0
9	49.0	39.0	61.0	0.0
10	66.0	55.0	69.0	1.0

To begin, we first obtain a plot of `verbal` against `quant`. To do this, we will extract `verbal` and name the new variable `"y"` and also extract `quant` and name that variable `x`:

```

y = df["verbal"]
x = df["quant"]
import numpy as np
import matplotlib.pyplot as plt
plt.scatter(x, y)
Out[327]: <matplotlib.collections.PathCollection at 0x1f4057b8>

```



We notice a somewhat moderate but linear relationship between the two variables. That is, an increase in `verbal`, on average, seems to be paired with an increase in `quant`. Of course, the plot is very bare, and so it would be nice to add some labels to the respective axes. This can be accomplished using the following (updated plot not shown):

```
plt.title("Scatterplot of Verbal on Quant")
plt.xlabel("Quant")
plt.ylabel("Verbal")
```

In Python, we will now run the regression of `verbal` on `quant`. When we say “verbal ON quant,” we mean `verbal` is the response variable and `quant` is the predictor. First, we add a constant to our predictor variable using `statsmodels.api`:

```
import statsmodels.api as sm
x = sm.add_constant(x)
```

The constant in the regression equation is the same for everyone, which is a value of “1” and may be readily verified (we print only the first three cases):

```
x
Out[30]:
      const  quant
0         1.0   56.0
1         1.0   42.0
2         1.0   43.0
```

We are now ready to run the regression using `sm.OLS`, which stands for “ordinary least-squares”:

```
model = sm.OLS(y, x).fit()
print_model = model.summary()
print(print_model)
```

```

                        OLS Regression Results
=====
Dep. Variable:          verbal    R-squared:                0.653
Model:                  OLS      Adj. R-squared:           0.641
Method:                 Least Squares    F-statistic:            52.68
Date:                  Tue, 09 Jun 2020    Prob (F-statistic):      6.69e-08
Time:                  14:02:26    Log-Likelihood:         -103.07
No. Observations:      30    AIC:                     210.1
Df Residuals:          28    BIC:                     213.0
Df Model:              1
Covariance Type:       nonrobust
=====
                        coef    std err          t      p>|t|      [0.025    0.975]
-----
const          35.1177         5.391      6.514   0.000     24.074     46.162
quant           0.5651         0.078      7.258   0.000         0.406         0.725
=====
Omnibus:                 1.230    Durbin-Watson:           1.244
Prob(Omnibus):           0.541    Jarque-Bera (JB):         1.020
Skew:                    0.431    Prob(JB):                 0.600
Kurtosis:                2.732    Cond. No.                 263.
=====
Warnings:
[1] Standard Errors assume that the covariance matrix of the errors
is correctly specified.
```


There is a lot going on in the above output, with enough statistical theory underlying its output to easily fill an entire book on regression. We summarize only the most salient and important features of the output:

- We confirm that the dependent variable (`Dep. Variable`) is that of `verbal` and that the model was estimated by ordinary least squares. The number of observations in the data is equal to 30. The degrees of freedom for `residual` are equal to 28, computed as the total sample size minus the number of predictors (which in our case is 1), minus 1. That is, $n - 1 - 1 = 30 - 1 - 1 = 28$. The degrees of freedom for the model (`Df Model`) are equal to 1, since there is only a single predictor. Had there been two predictors, this number would equal 2.
- The `R-squared` for the model is equal to 0.653, and is the proportion of variance in the response accounted for by the predictor. That is, knowledge of `quant` accounts for approximately 65% of the variance in `verbal`. Adjusted R-squared is equal to 0.641 and, as expected, is slightly less than R-squared. However, since the model is not overly complex (we only have a single predictor, after all), Adj. R-squared is essentially almost equal to R-squared.
- The *F*-statistic for the model is equal to 52.68, with an associated probability of 6.69e-08, which translates to a *p*-value of 0.0000000669. Hence, the model is easily statistically significant, which indicates that the predictor is affording prediction of the response over and above chance. As we will see, since there is only a single predictor in the model, this *p*-value will agree with the *p*-value obtained for the *t*-statistic on `quant` that we will survey shortly.
- **AIC** for the model is equal to 210.1. We see that the **BIC** value is only slightly larger. AIC and BIC, like Adjusted R-squared, are penalized fit statistics. Though we do not discuss them in detail in this book, for applied purposes, you should know that **smaller values than not are preferred**, as both of these measures penalize for “too complex” of a model. Typically, BIC penalizes a bit more harshly so we expect it to be a bit larger compared to AIC.
- Next in the output we come to the coefficients for the model. We see that the intercept (`const`) is equal to 35.1177 with standard error equal to 5.391. When we divide 35.1177 by 5.391, we obtain the *t*-statistic of 6.514. We see that it is statistically significant ($p = 0.000$), with confidence limits of 24.074 and 46.162. Recall what the constant represents; it is the predicted value for the response when the value of the predictor variable is equal to 0, since $y_i = a + bx_i = a + b(0) = a$. It usually is not of immediate interest, in that we are much more interested in the value for the coefficient for `quant`, the predictor in the model.
- The coefficient for `quant` is equal to 0.5651, and we see it is also statistically significant ($p = 0.000$). This is the slope for the regression equation and is interpreted as: **for a one-unit increase in `quant`, we expect, on average, a 0.5651 unit increase in `verbal`, given the model under test**. Now, the model “under test” is a simple linear regression, but this statement is still required. When we study multiple regression shortly, we will see that when we include additional predictors into the model, the coefficients for other predictors typically change. Hence, “given the model under test” is a statement you should include regardless of the complexity or simplicity of the model. **Any effect in a model is always evaluated relative to the model, never in a vacuum.**

- The remaining pieces of the output are not of immediate concern and hence we do not interpret them here.

This completes our discussion of simple linear regression, and we now move on to consider the more popular multiple regression model. As we will soon see, interpreting multiple regressions, while having some similarities to simple regression, includes additional complexities. Understanding these complexities is very important. Interpreting effects in a multiple regression is not the same as interpreting them in a simple regression, and we will discuss why this is so in some detail.

7.10 Multiple Linear Regression

In the simple linear regression just conducted, we only had a single predictor. Often, however, a researcher wishes to model the relationship between several predictors and a single response simultaneously (recall we included simultaneous independent variables in factorial ANOVA as well). When we include more than a single predictor into a regression model, the model is that of **multiple linear regression**. Technically, what we are doing is modeling the response as a function of a **linear combination** of predictor variables.

At first glance, you may think multiple regression is a simple extension of simple linear regression, and whether a researcher chooses to do one or the other is of little consequence. This impression, however, is misguided. When one performs a multiple regression, there are numerous issues that arise that simply did not exist in simple linear regression models, or, if they did exist, were not as apparent and “in your face” as for multiple regression models. In multiple regression, we cannot avoid those issues and must instead confront them head on.

7.11 Defining the Multiple Regression Model

Where the simple linear regression model was defined as

$$y_i = \alpha + \beta x_i + \varepsilon_i$$

the multiple linear regression model is given by

$$y_i = \alpha + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_p x_{pi} + \varepsilon_i$$

A few things to note when comparing the two models:

- Both models feature only a **single response variable**, y_i . That is, whether one performs a simple linear regression or a multiple linear regression, there is still only a single response. If there were more than a single response, the model would be **multivariate** in nature.
- There is only a single intercept, α , in both models.
- Whereas the simple linear regression model features one predictor with an associated regression weight, βx_i , the multiple regression model features several predictors, each

with associated regression weight, $\beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_p x_{pi}$. While the weight in simple linear regression is simply a coefficient for x_i , the weights in multiple regression, because there are several of them, are best referred to as **partial regression weights** for each predictor x_{1i} through to x_{pi} . Notice that each predictor variable is still indexed by an “ i ” following p , to indicate the i th value of the given predictor.

- Both models, whether simple or multiple regression, have an **error term** associated with them, ε_i . The error term is still subscripted by a single “ i ” in the multiple regression model as it was in the simple regression model, since there is only a single error possible regardless of how many predictors one has in the model. This is important to understand. Simply because we have included more than a single predictor does not mean we are no longer obtaining a single prediction of the response. We are still obtaining a single prediction, only that the response we are predicting is based on a more extensive linear combination of predictor variables. Hence, there is still only a single error term, which implies there will be only a single **residual** of the form $y_i - \hat{y}_i$ as well.

It cannot be emphasized enough how multiple regression is different from simple regression, and in this introduction to multiple regression, it allows us to introduce some of the primary features of statistical modeling in general. We have already mentioned it in passing, but now it is time to name it as a fundamental principle of statistical modeling. The following is what we like to call the **universal principle of statistical modeling**, no matter how simple or complex the model:

All models are context-dependent.

It would not be an exaggeration to say that every research paper that features statistical modeling should begin their report of their statistical results by including the above sentence. **Context-dependency** is the idea that whatever model you evaluate, it is always the **model** you are testing, not individual predictors within that model. For example, if I set up a model to predict depression, and include anxiety and social isolation, the regression weight obtained on both anxiety and social isolation cannot be interpreted separately from each other. That is, if a regression weight of 0.70 is obtained for anxiety, that coefficient should only be interpreted in the **context** of having social isolation also in the model, and not distinct from it. This is equally true for the effect of social isolation, in that it can only be interpreted in the context of anxiety. So, if anxiety is statistically significant, that statistical significance “exists” in the context of also having included social isolation, and not **necessarily** distinct from it. Likewise, the obtained estimate of the regression weight is also context-dependent. That is, it exists in the context of the other variable(s). The regression estimates are not independent of the model tested and do not exist in a vacuum from it. For instance, if we only included anxiety in the model, the coefficient for anxiety might change from what it was when social isolation was also included. If a researcher or consumer of research is unaware of this context-dependency of the model he or she is generating or consuming, then there lies a serious risk of misinterpretation, thinking that anxiety is a predictor of depression, **period**. This may not be the case. Anxiety is a predictor given the context in which its coefficient is estimated. Never forget that!

How can we solve the context-dependency issue? Given our discussion above, you may at first attempt to argue that if you want to avoid the issue, simply conduct simple linear regressions and not multiple. However, this idea is misguided because even in a simple regression, **you are still testing the model!** The fact that you are excluding predictors from your model is still an act of making your model context-dependent, only that in this case, you are not including variables that may be relevant to the model. In brief, **you cannot escape the issue!** It will haunt you regardless of the model you fit, even if very simple. Statistical models do not necessarily reflect reality, even if they are often advertised to reflect just that. They partition variability and estimate parameters, that is all. The rest of the substantive conclusions must come from **good research design** and/or quality variables that you are subjecting to the regression in the first place.

7.12 Model Specification Error

The idea of context-dependency is allied to the idea of specifying the correct model. All statistical models, including regression, assume that what you are fitting is a model that includes the most relevant predictors to the response. If the model you are fitting is grossly incomplete, such that important and influential covariates are lacking from the model, then not only will the parameter estimates featured in the model be incorrect, any conclusions about the predictive power of the model will likewise be wrong. Of course, very few, if any models can be perfectly specified, which is in part why George Box infamously once said, “**All models are wrong, some are useful.**” The problem of **omitted variables** is sure to plague any regression model you fit, even if only on a theoretical level in some cases. For instance, perhaps a given treatment is a strong predictor of COVID-19 recovery, but there will always be more variables that could be added to the model to help improve on its **specification**. That does not imply that we should not take the COVID-19 regression model seriously that was fit, only that we appreciate the fact that it is still an **incomplete model**. The model you fit will be wrong, even if it fits the data perfectly. Don’t feel bad, as Newton’s models were wrong too, even if they explained (and still do) a great deal of variability. **Science (if done ethically) is a very tough game.** Be prepared to be wrong all the time! However, if you can explain variability and perhaps theorize on causal mechanisms, you are doing pretty well! Don’t despair!

The moral of the story is this – when you fit a regression model, you should always try to ensure it is as well-specified as possible. Otherwise, parameter estimates in the model and conclusions drawn about the model effects will be biased and wrong. When interpreting any statistical model in the literature, the first critical question you should ask yourself when reading the article is whether the model is **correctly specified**. If it is not, or the author does not address the specification of the model in some detail, then the resulting model may not be worth interpreting at all, as it can potentially misguide more than it guides. A good research paper should, in some sense, address the model specification issue. In the physical and biological sciences, the goodness of the specification is usually implicit and requires less backing. In the social sciences, however, such as with economic or psychological variables, model specification should be rigorously defended before the statistical analyses are carried out and interpreted.

7.13 Multiple Regression in Python

We now demonstrate a multiple regression, this time using both `quant` and `analytic` simultaneously as predictors of `verbal`. In this example, we refrain from interpreting *p*-values and simply focus on obtaining the relevant coefficients. Recall our IQ data:

```
import pyreadstat
df, meta = pyreadstat.read_sav("iq_data.sav")

df
Out[169]:
```

	verbal	quant	analytic	group
0	56.0	56.0	59.0	0.0
1	59.0	42.0	54.0	0.0
2	62.0	43.0	52.0	0.0

First we import the required packages:

```
import pandas as pd
from sklearn import linear_model
import statsmodels.api as sm
```

We then bin together the predictor variables along with the response. Note that `X` is now composed of both `quant` and `analytic`:

```
X = df[['quant', 'analytic']]
Y = df['verbal']
```

Next, we define the regression object:

```
regr = linear_model.LinearRegression()
regr.fit(X, Y)
Out[180]: LinearRegression(copy_X=True, fit_intercept=True, n_
jobs=None, normalize=False)
```

At this point, the regression has been performed and we are ready to extract parameters from it. We now extract the intercept fit by the model:

```
print('Intercept: \n', regr.intercept_)
Intercept:
32.17094152928748
```

We see that the intercept for the regression equation is equal to 32.17. Next, we obtain the partial regression coefficients, the first for `quant`, the second for `analytic`:

```
print('Coefficients: \n', regr.coef_)
Coefficients:
[0.42814931 0.16964027]
```

The coefficient for `quant` is equal to 0.428, while the coefficient for `analytic` is equal to 0.169. How each of these is interpreted is very important. At first glance, it may be tempting to interpret `quant` in this way:

For a one-unit increase in `quant`, we can expect, on average, verbal to increase by 0.428 units.

This interpretation is wrong! But how can this interpretation be incorrect? It sounds so consistent with how we interpreted the regression slope in simple linear regression. And therein lies the problem! **The nature of the coefficient is different in multiple regression than it was in the simple linear regression.** That is, we cannot interpret it in the same way. Since the coefficient appears in the context of another coefficient, it is imperative in our interpretation to provide the “context” statement that we initially provided in the simple linear regression. There are two ways of doing this. Either one of the following two interpretations would work:

For a one-unit increase in quant, we can expect, on average, verbal to increase by 0.428 units, given that the model under test has analytic also in the model.

For a one-unit increase in quant, we can expect, on average, verbal to increase by 0.428 units when holding analytic constant.

Either one of these statements will adequately describe the model results. Notice that in both statements, the context is given for the model. Without either of these, the interpretation is literally wrong.

We can obtain predicted values for our model by designating specific values for `quant` and `analytic`. For instance, suppose our new input for `quant` is equal to 20, while our new input for `analytic` is equal to 25:

```
new_quant = 20
new_analytic = 25
print('predicted verbal: \n', regr.predict([[new_quant, new_
analytic]]))

predicted verbal:
[44.97493447]
```

We see that given the above inputs, our predicted value for `verbal` is equal to 44.97.

7.14 Model-Building Strategies: Forward, Backward, Stepwise

In this section, we survey approaches to model-building in regression. What does it mean to “build a model?” There are essentially two steps to model-building: (1) determining the set of predictors you will consider to be candidate predictors for the model and (2) adopting a method by which those predictors will be entered into the model. Despite all of the statistical advances and sophistication in model-building over the years, and many algorithms and such developed to aid in the process, the most important step to model-building remains choosing candidate predictors for consideration into the model. A **candidate predictor** is a predictor you have measured or obtained and will consider it for possible and potential inclusion into a model you are building. Without the careful selection of these candidate predictors, any model you build will necessarily be problematic and possibly very misleading. Hence, this necessarily presents a problem:

There is no statistical method that will ensure you have selected the correct or most complete set of candidate predictors.

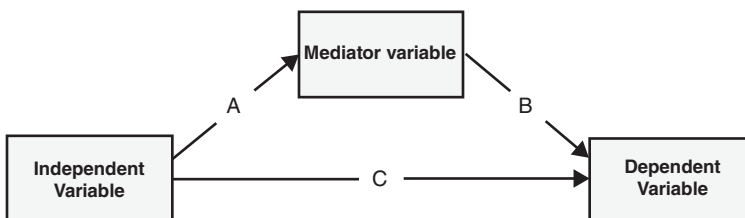
That is, sophisticated as model-building has become over the last century and despite all of the advances in software, the first step to model-building is not a statistical one at all! Rather, it is a **scientific** one. No statistical algorithm can ensure you have considered the most useful and relevant of predictors. You, as a scientist, must be “on top” of this from the start. For example, for the case of predicting COVID-19 illness, suppose that a researcher designed a model in which pre-existing conditions as a predictor in the model were not considered. The failure to even record this information or include it into the model would constitute a major **specification error**, and the model would be wholly incorrect regardless of which model-building algorithm was then used. If the set of candidate predictors is wrong, or incomplete, using any technology afterward to build the “best” model will simply not work. Hence, as a scientist, you must ensure that before you build a model, you are including the most complete and accurate set of predictors possible before you set the algorithm into motion. That is, **the selection of candidate predictors is the most important step in all of statistical modeling building**. It is also the least technical. Once you have selected your candidate predictors, the second step is consideration on how to enter them into your model or how to designate an algorithm to search for the most “important” (in some sense) predictors. Of these, there are several options: simultaneous, hierarchical, forward, backward, and stepwise regression (among others). We briefly consider each of these.

In **simultaneous regression**, otherwise known as **full entry** regression, the researcher, after determining a suitable set of candidate predictors, enters all of these predictors into the model at once. This type of regression should be considered your “default” choice in most of your data analyses. That is, unless you have a good reason for not entering all of your predictors at once, then most regressions you perform should follow this approach. Sometimes researchers like to preselect the predictors they enter into their model, and enter each individually or as a group at each stage of the model-building process. This is known as the **hierarchical approach** to model-building. For instance, suppose we already know that pre-existing heart conditions can predispose one to COVID-19. However, we would like to know over and above this whether gender is an important predictor. We could build the model by first entering a variable representing pre-existing heart conditions (to keep it simple, though definitely likely medically incorrect, suppose we are entering beats-per-minute), then enter gender as the next predictor. When we examine the effect of gender, we are studying the predictive impact of gender over and above that of pre-existing heart conditions. Hence, the hierarchical model gives us a convenient way to observe the variance explained by adding predictors to our model. It needs to be recognized, however, that regardless of whether you enter predictors in a hierarchical fashion or full entry, the final model results will be the same. That is, to use our example, whether you first enter pre-existing conditions or include them directly with gender, the final model will give you the same results. Hierarchical is preferred, however, in some cases where the researcher would like to observe and take account of the **incremental variance explained** by adding one or more predictors.

The idea of adding predictors into a model after others have already been included is sometimes referred to as **controlling for other predictors**. However, it needs to be emphasized that this is **statistical control** only, not **experimental**. There is no true control unless you have an experimental design that features such control. All we are

doing when we observe the effect of one variable in the presence of others is partialling out variability, nothing more. Statistical control does not somehow serve as a panacea to experimental control, nor is it a justifiable way to get out of performing rigorous experiments. Beware of misinterpretations of statistical control. Statistical control is not that big of a thing, scientifically speaking.

Hierarchical regressions form the basis for a procedure some researchers sometimes perform, that of **statistical mediation**. In a mediation model, as shown in the following diagram, an independent variable is hypothesized to predict a response variable, but such a relationship is said to be “mediated” by another variable (the so-called “**mediator**”). As the story goes in mediation, if the original relationship between the IV and DV disappears, or nearly so, then varying degrees of support for mediation are said to exist.



Of course, the establishment of statistical mediation in no way justifies, on its own, a conclusion of a **physical process** between the IV, mediator, and DV, since as we can clearly see, statistical mediation is, at most, simply an example of hierarchical regression. Now, if a substantive argument can be advanced that “true mediation” on a physical substantive level is occurring, then statistical mediation might serve as support for that argument. Too often, however, researchers falsely believe that the establishment of statistical mediation implies or somehow justifies a physical or scientific “process” between variables. This is grossly incorrect. **No statistical model, on its own, can justify any process that does not exist before the statistical model was applied.** Still, and even if regularly misunderstood, mediation analysis is quite common, especially in psychology and related areas. If the **a priori assumptions** governing the variables are tenable, mediation analysis can indeed be quite useful on a scientific level. But if such assumptions are not tenable, then all that is being performed for certain is a hierarchical regression (i.e. variance partitioning) and possibly nothing more. Unless of course there is an underlying process we are simply not aware of yet, but it would still be a gigantic leap to say the independent variable “goes through” the mediator to the response in any sense. All we can really say is that the mediator “accounts for” the relationship between the predictor and response variable, but given that many variables can “account for” bivariate linear relationships, the substantive conclusion may simply still not be that impressive if not strongly governed by theory or substantive context (e.g. working with biological or physical variables). In a real sense, all correlations are “spurious,” which implies they are “due” to one or more other variables, but if you are performing correlational analyses, you can only conclude this in a correlational sense. An experiment would be required to detect any true “due to a third variable” idea and attribute any **causal** or **impact** mechanism. Hence, the researcher that concludes, “We found evidence for a mediating relationship,” may simply have found evidence that variance can be successfully partitioned among variables, which is not all that ground-breaking. For extensive instruction on mediational analyses, see MacKinnon (2008).

7.15 Computer-Intensive “Algorithmic” Approaches

Thanks to advances in computing power in the last 40–50 years, algorithmic approaches to model selection have become quite popular. Now, let us be sure to understand what this means. A computer does nothing more than obey the commands it is instructed with. Hence, saying that a computer is “performing” model selection for you is a hollow and meaningless statement. What is choosing the model is the code and instructions the user has programmed into the software. This code and instructions are based on, you guessed it, “anticlimactic” (i.e. it won’t make cable news) mathematics. Hence, if you want to truly understand algorithmic approaches to model selection, **you need to study the underlying machinery**. You need to understand the sticks and stones. You need to understand what the computer is actually “doing” when it is selecting the model. Of course, we could just test every conceivable model to see which fits best, so-called **best subset regression**, but for p predictors, that would entail testing 2^p models. For instance, for the case of $p = 30$ predictors, the number of models possible is equal to $2^p = 2^{30} = 1,073,741,824$. Choosing to adopt an algorithmic approach to model building may help, and “shrink” the task down to something more manageable, though in managing complexity, there is certain to be an associated cost. Whenever you attempt to trim a problem down to a manageable size, you always lose information along the way.

In **forward regression**, the algorithm searches among the candidate predictors (remember, it cannot identify predictors you do not introduce for consideration to begin with) and selects that which has the largest **bivariate correlation** with the response at some pre-designated alpha level, such as $\alpha = 0.05$. After selecting this first predictor, the model searches among remaining predictors for the one having the highest **squared semipartial correlation** with the response, which essentially translates to the predictor having the greatest statistically significant increase in variance explained in terms of an F -statistic. The algorithm continues in this fashion until no other predictors meet entrance criteria. Key to the forward algorithm is that **it will not remove predictors**. That is, **once a predictor is in the model, it remains in the model**. As we will see, this is contrary to stepwise regression in which predictors are allowed to be removed at each step. In **backward elimination**, the process is reversed, such that the model starts out with all candidate predictors and removes at each step predictors that contribute the least to the model.

In **stepwise regression**, the model first searches for predictors for inclusion in the spirit of forward regression, but then at each step also considers predictors for removal. For example, once the second predictor is entered, the first predictor is re-assessed for statistical significance given the inclusion of the second predictor. If it is no longer statistically significant, it is removed from the model. Stepwise continues in this fashion until no predictors are entered or removed. While the entrance criteria is typically set at an alpha level of 0.05 or so, the removal criteria must be at a level greater than this (e.g. 0.10); otherwise, the algorithm could engage in cyclic behavior of adding and removing predictors without any progress to model fit.

7.16 Which Approach Should You Adopt?

The decision of which model-selection procedure to adopt is thus not an easy one, and as a researcher, you need to be well aware that a given approach may yield different results than adopting a competing approach. Does this problem sound familiar? It

should. It shows once again that the model you fit and interpret is contingent upon decisions made along the way. Not only then is model fit determined by what predictors you do or do not include, but it also depends on the methodology used in model selection. A stepwise approach may yield an entirely different model than a forward or simultaneous regression, and the researcher generating such research as well as the consumer consuming such research needs to be intimately familiar with these issues. Otherwise, they are likely to mistake a given model as “the” model that represents the data. It is never “the” model. Rather, it is a model determined by all the inputs we are discussing. **Subjectivity is part of the model-building process.** The moment you decide to build a model, you will, to some extent, be making subjective decisions. So long as you are clear and upfront about these decisions, that is fine, but you need to communicate them to your audience to ensure you are being **authentic** and **ethical**.

One should also be aware that though stepwise approaches may appear attractive at first, they come with their own set of difficulties. For one thing, parameter estimates are usually biased when using an incremental F -test as used in such approaches (Izenman, 2008). Hence, computed standard errors may not be entirely realistic, leading the model to include or not include given predictors. Among students, stepwise regression is often a “first choice” when deciding on a model-building algorithm due to its perceived complexity and “coolness” factor (everyone wants to do complex things), but it is usually not the best choice. The best choice is usually the anticlimactic simultaneous regression where decisions about predictors are made based on a researcher’s knowledge of the literature, along with hypothesized outcomes. As with many things in statistics, **it is often the simpler tool that is the most effective**, and hence unless your research is nothing more than a fishing expedition, simultaneous regression will usually meet your needs.

Utility should also be at the forefront of what predictors to include. If a predictor is too expensive to recruit from subjects (for instance, maybe some blood chemistry tests are exceedingly expensive to obtain), then you may be better off going with a simpler predictor that accounts for as much or nearly as much variance in the response variable. A model explaining slightly less variance may nonetheless be a “better” model if its candidate predictors are more easily obtained. If it were found that heart disease could be predicted by rocketing people into space and observing circulatory behavior in microgravity, strong as a predictor this may be (I’m just making this up), it would have virtually zero utility as a predictor (here on planet Earth) since obtaining that information on a sample of individuals would be virtually impossible, if not extremely expensive. **Good predictors explain variance, but they are also manageable, obtainable, and “do-able” for the project. That is, good predictors have good utility.** Once you bring modeling down from the ivory tower of abstractness, these “pragmatic” things matter a great deal. Try convincing a top business CEO of the most “theoretically correct” model and you may be surprised at the response you get! Pragmatics matter!

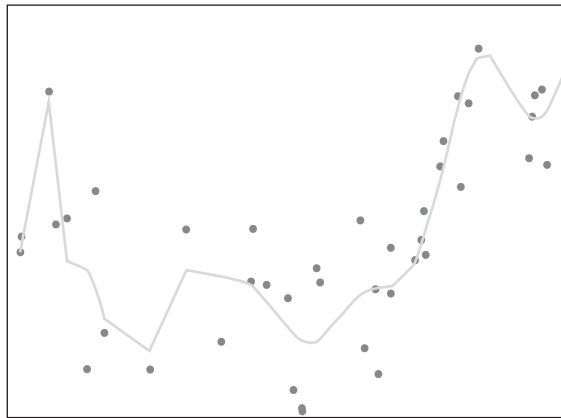
7.17 Concluding Remarks and Further Directions: Polynomial Regression

In this chapter we have merely scratched the surface on the topic of regression analysis. Regression can easily take up a full year course, and even then we would still be only scratching the surface! For an excellent and thorough treatment of all things about

regression, as well as extensions to the regression model into the generalized linear model and all its possibilities, see Fox (2016). Draper and Smith (1998) is also a classic reference and should be consulted. Denis (2021) also features a more thorough chapter on both simple and multiple regression, as well as interactions within a regression framework.

In this chapter, we have assumed that relationships among response variables and predictors could be modeled sufficiently as a **linear** trend. In many cases, however, **non-linear** trends may provide a better fit, both statistically and in line with one's substantive theory. In those cases, **polynomial regression** may be a suitable option, where one can flexibly change the degree of the polynomial away from degree 1 to learn what best fits the data. This chapter has been about first-degree polynomials (otherwise known as “lines”); that is, $y_i = a + bx_i + e$ featured an input variable x_i understood to be raised to the first power. An example of a more complex polynomial is that of $y_i = a + b_1x_i + b_2x_i^2 + e$, where now the predictor is raised to a higher exponent, in this case “2,” making the model a **quadratic** one. Such models are still considered **linear in the parameters** since b is still to the first power. However, the curve of these models is different and may necessitate fitting a different polynomial other than a linear one.

From a statistical and computational point of view, fitting polynomial models is not too difficult, and is quite similar to fitting ordinary linear regression models. However, from a scientific point of view, fitting polynomial models can quite easily be **abused** as the analyst attempts to account for as many data points in the plot as possible, thereby potentially “**overfitting**” the model to the data. As an example, consider the following plot:



This is an example of overfitting a model to data, since the curve is attempting to essentially account for as many data points as possible. What could possibly be wrong with this? Isn't the goal of modeling to best account for the data? While from a statistical point of view the curve might fit quite nicely, from a scientific point of view it is unlikely that a replication of the study or validation of the model on new data (i.e. “cross-validation”) would generate such a tight model fit. Without knowing the material under investigation in the plot (i.e. without knowing the “stuff” on which the science is being performed, that is, the raw objects or material), one cannot adequately assess the extent of the overfit, at least not on an intuitive level. Statistically, one could separate the data into a **train** and **test** set, and attempt to validate the model to see

how well the curve fits on the test data. Slightly “**under-fitting**” a model may be preferred to “over-fitting,” especially from a scientific perspective, so that whatever model is fit, is fit with a bit more **generalizability** to new data. By way of analogy, if you are a clinical psychologist, over-fitting your diagnosis to a particular client makes it a **case study**, not a generalizable phenomenon. Again, these issues cannot truly be understood without recourse to the scientific material under investigation. In sum, anyone can “connect the dots” on a plot, but that is not what mathematical or statistical modeling is about, just like explaining why one individual contracted COVID-19 is hardly an explanation for what causes the disease. **Generalizability**, as well as **simplicity** are always the goals, not to fit a ridiculous model to demonstrate a failure to relate the statistics you are computing to the scientific data you are modeling. For further details on polynomial regression, James et al. (2013) is an excellent introduction.

Review Exercises

1. Discuss the goal of **regression analysis**. Is it to make predictions? If not, what is it for?
2. Given an example (not featured in the chapter) in the age of COVID-19 where a **simple linear regression** may prove useful.
3. What is the **least-squares** principle? What does least-squares regression guarantee? What can it not guarantee?
4. What does it mean to **estimate parameters** in regression? Under what circumstance would parameter estimation not be required? Is such a circumstance realistic? Why or why not?
5. Consider the formula for the **slope** coefficient for the regression of y_i on x_i :

$$b_{y.x} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Relying on your knowledge and intuition behind what the formula is accomplishing, rewrite the formula for the slope where x_i **is regressed on** y_i instead. What changes would you make to the equation and why? Why do such changes make good sense in terms of what regression analysis accomplishes?

6. Distinguish between an **error** and a **residual** in regression analysis. Why are they not one and the same thing? Do we ever have access to errors? Why or why not?
7. Why is it the case that the **coefficient of determination**, or R^2 , cannot typically exceed a value of 1.0 in either simple or multiple linear regression? Justify that it cannot.
8. Your colleague, after conducting her research study, tells you, “The coefficient obtained in my regression is equal to 0.90.” Would you consider this an impressive result? Why or why not? What other questions might you ask her?
9. For a constant value of R^2 , what happens to the size of R^2_{Adj} as p increases? What about as n increases? Justify your argument using the formula in each case.
10. Consider the **Forest Fires Data Set** from the **Machine Learning Repository** (<https://archive.ics.uci.edu/ml/index.php>). Regress area burned on the temperature

variable in Python. Is temperature a good predictor of how much land will be burned in a fire? How much variance is explained?

11. What is the effect of lowering the estimated **standard error** for a predictor in regression on the **p-value** associated with the resulting *t*-statistic? All else being equal, are small standard errors a good thing or a bad thing for your research project? Why?
12. Consider the **Challenger USA Space Shuttle O-Ring** data set housed at the **Machine Learning Repository**. You can learn more about the data by visiting the site. Regress the launch temperature (degrees F) on leak-check pressure (psi) and comment on the predictive relationship, if any.
13. What is meant by the statement “**All models are context-dependent**” and why this is possibly the most important take-away from learning statistical models from a scientific point of view? Unpack and discuss the implications of this statement and why it is so important.
14. Discuss the difference between **hierarchical** and **stepwise regression**. How are they not the same? Is stepwise regression ever appropriate? If so, when?
15. Why are **mediational analyses**, especially in questionable scientific contexts, a potential philosophical quicksand when it comes to interpretation? Discuss.